

Day 26: SQL Introduction & Core Concepts

1. WHAT IS SQL & BASIC PARADIGM

SQL (Structured Query Language) is a domain-specific language used to communicate with, manage, and perform operations on structured data stored in *Relational Database Management Systems (RDBMS)*.

Basic Flow Architecture
User / Application → SQL Interface → Database Engine (RDBMS)

SQL is utilized across seven fundamental data management operations:

- Creating and modifying database structures (Schemas, Tables)
- Inserting new records into relations
- Retrieving data based on specific filtering rules
- Updating existing values in tables
- Deleting obsolete data records
- Enforcing rules and integrity constraints
- Controlling access control and user privileges

2. WHY "STRUCTURED" QUERY LANGUAGE?

In relational databases, data is organized strictly into **Tables (Relations)** consisting of predefined rows (tuples) and columns (attributes).

Student_ID	Name	City
101	Ravi	Delhi
102	Aman	Mumbai

Because structural parameters (datatypes, relationships, keys) govern how data is organized, SQL acts as the standard declarative tool for querying structured relational data.

3. HISTORY & EVOLUTION OF SQL ★

Understanding SQL's historical lineage is critical for competitive exams and interviews:

- **1970 — Relational Model:** Proposed by **E. F. Codd**. It relies heavily on mathematical foundations: *Relational Algebra* and *Tuple Relational Calculus*.
- **IBM Implementation:** IBM practically implemented Codd's model and introduced **SEQUEL** (*Structured English Query Language*), which was later shortened to **SQL**.
- **Industry Adoption:** Tech giants developed major commercial engines around SQL including *Oracle*, *Microsoft SQL Server*, *IBM DB2*, and *MySQL*.

 **Exam Memory Pipeline**
E. F. Codd → Relational Model → IBM / SEQUEL → SQL Standard

4. SQL IS A DOMAIN-SPECIFIC LANGUAGE (DSL) ★

A **Domain-Specific Language** is built specifically for a target problem domain, unlike general-purpose languages.

Language	Type	Primary Purpose
C / C++ / Java	General-Purpose	System software, desktop applications, enterprise logic
SQL	Domain-Specific	Relational database queries and operations

6. SQL VS. PL/SQL

PL/SQL (Procedural Language extensions to SQL) is an Oracle-specific extension that introduces procedural capabilities around pure SQL.

SQL

- Purely Declarative language
- Executes single commands independently
- No native control flow (loops, branches)

PL/SQL Extension

- Procedural extension (Oracle)
- Supports procedures, functions, triggers
- Adds cursors, variables, loops, exception blocks

⚠ Clarification: Standard SQL remains declarative; adding procedural wrappers around SQL (like PL/SQL) does not make the underlying SQL query language procedural.

7. MAIN CATEGORIES OF SQL COMMANDS ★★★

Category	Full Name	Primary Purpose	Key Commands
DDL	Data Definition Language	Defines or alters database structures/schema	CREATE, ALTER, DROP, TRUNCATE, RENAME
DML	Data Manipulation Language	Manipulates data stored inside tables	INSERT, UPDATE, DELETE
DQL	Data Query Language	Retrieves structured data from relations	SELECT
DCL	Data Control Language	Manages user privileges and permissions	GRANT, REVOKE
TCL	Transaction Control Language	Manages database transaction boundaries	COMMIT, ROLLBACK, SAVEPOINT

8. KEYS AND INTEGRITY CONSTRAINTS

Constraints enforce business rules, consistency, and structural integrity across relational data tables.

- **PRIMARY KEY:** Uniquely identifies each record in a table. Disallows NULL values.
- **FOREIGN KEY:** Maintains referential integrity between child and parent tables.
- **UNIQUE:** Guarantees all non-null values in a column are distinct.
- **NOT NULL:** Ensures a column cannot hold missing/NULL entries.
- **DEFAULT:** Automatically populates a specified default value when none is supplied.

```
CREATE TABLE Student (  
    Student_ID INT PRIMARY KEY,  
    Name VARCHAR(50) NOT NULL,  
    City VARCHAR(30) DEFAULT 'Koppal'  
);
```

9. CORE SQL OPERATORS ★

Specialized operators streamline complex database queries within the WHERE clause:

Operator	Description	Example Usage
BETWEEN	Filters values within an inclusive range	WHERE Marks BETWEEN 60 AND 80;
IN	Matches against discrete list of values	WHERE City IN ('Delhi', 'Mumbai');
NOT IN	Excludes values in discrete list	WHERE City NOT IN ('Delhi', 'Mumbai');
EXISTS	Returns TRUE if subquery returns ≥1 row	WHERE EXISTS (SELECT 1 FROM ...);

10. ORDER OF SQL CLAUSES ★★ ★

Mastering both the written order and execution sequence of clauses is vital for writing accurate queries:

```
SELECT City, COUNT(*)
FROM Student
WHERE Marks >= 50
GROUP BY City
HAVING COUNT(*) > 2
ORDER BY City;
```

🧠 Memory Chain — Standard Order

FROM → WHERE → GROUP BY → HAVING → SELECT → ORDER BY

11. AGGREGATE FUNCTIONS

Aggregate functions execute operations on a set of values and return a single scalar summary value:

- MAX(): Returns highest value in column set.
- MIN(): Returns lowest value in column set.
- AVG(): Computes average numeric value (ignores NULLs).
- COUNT(): Counts rows/entries matching specified condition.
- SUM(): Calculates total summation of numeric attributes.

12. JOINS & SUBQUERIES ★★ ★

Joins: Combine records from multiple relational tables based on logically related fields.

[Student Table] + [Course Table] → (JOIN) → Combined Enrolment Report

Subqueries (Nested Queries): An inner query nested within an outer query construct.

```
SELECT Name FROM Student
WHERE Marks > (SELECT AVG(Marks) FROM Student);
```

Correlated Subqueries refer back to columns in the outer query and execute iteratively row-by-row.

13. PRACTICAL REAL-WORLD EXAMPLE: API INTEGRATION (IRCTC)

Modern applications consume backend SQL infrastructure via backend service APIs:

User Inputs: Date & Station → API Request → RDBMS SQL Execution → Seat Availability Output

14. HIGH-PRIORITY REVISION MIND MAP

SQL Quick Architecture Overview | — Relational Data (Tables, Relations, Rows/Columns) | — Domain-Specific & Declarative (Specifies WHAT, not HOW) | — Command Subsets | — DDL : CREATE, ALTER, DROP, TRUNCATE | — DML : INSERT, UPDATE, DELETE | — DQL : SELECT | — DCL : GRANT, REVOKE | — TCL : COMMIT, ROLLBACK, SAVEPOINT | — Core Filtering & Logic | — Operators : IN, BETWEEN, EXISTS, NOT IN | — Clauses : FROM -> WHERE -> GROUP BY -> HAVING -> ORDER BY | — Functions : MAX, MIN, AVG, COUNT, SUM | — Advanced Placement Topics (High Weightage) | — JOINS (Inner, Outer, Self, Cross) | — Subqueries / Nested Queries & Correlated Queries | — PL/SQL (Procedures, Functions, Triggers, Cursors)

🧠 One-Line Exam Summary

SQL = Structured + Relational + Domain-Specific + Declarative + Database Query Language.